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Inventor(s)	ten Have; Eric Stephan

Wavelength division multiplexing device with passive alignment substrate

Abstract

A wavelength division multiplexing device includes an alignment substrate configured to provide alignment between optical components of the device. The device includes a plurality of collimating lenses, and the alignment substrate includes a plurality of aligners. Each of the aligners is configured to place a respective one of collimating lenses in a predetermined position and a predetermined orientation with respect to the other collimating lenses. The alignment substrate thereby provides passive alignment of the collimating lenses with a designed optical path. The substrate may also include visual alignment markings that provide an indication of the placement of multi-layer thin film filters so that the filters define an actual optical path in alignment with the designed optical path, and integrated optical waveguides that provide an optical beam to each of the collimating lenses.

Inventors: ten Have; Eric Stephan (Berlin, DE)

Applicant: CORNING RESEARCH & DEVELOPMENT CORPORATION (Corning, NY)

Family ID: 87850352

Assignee: CORNING RESEARCH & DEVELOPMENT CORPORATION (Corning, NY)

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/686,616, now U.S. Pat. No. 11,927,804 B2, filed on Mar. 4, 2022, the content of which is relied

upon and incorporated herein by reference in its entirety and the benefit of priority under 35 U.S.C. § 120 is hereby claimed.

TECHNICAL FIELD

(1) This disclosure relates generally to optical wavelength multiplexing, and more particularly to a wavelength division multiplexing device including passive alignment features that align one or more optical components of the device.

BACKGROUND

(2) Optical fibers are useful in a wide variety of applications, including the telecommunications industry for voice, video, and data transmissions. Benefits of optical fibers include wide bandwidth and low noise operation. However, the need to connect network nodes with fiber-optic cables often drives the cost of fiber-optic networks, especially for fiber-optic networks having a large geographic footprint.

(3) One way to increase the capacity of existing fiber-optic networks while avoiding the need to lay additional fiber-optic cables is through Wavelength Division Multiplexing (WDM). WDM involves transmitting data with multiple optical signals in a single optical fiber. Each optical signal has a different wavelength from the other optical signals. The single optical fiber transmits the different optical signals simultaneously in the same direction. Thus, the different optical signals provide different channels for data in the single optical fiber, and may be referred to as “signal components” or “optical carriers” of a combined optical signal or “optical beam” carried by the optical fiber. Because of their cost-effectiveness, WDM applications are increasingly being deployed to address the growing demand for bandwidth.

(4) WDM systems require hardware to combine and separate the different optical signals at various locations in an optical network. Typically, this hardware includes a WDM device (referred to as a multiplexer) that combines the individual optical signals into one optical beam at one network location, and another WDM device (referred to as a demultiplexer) that splits the optical beam into the individual optical signals at another network location. WDM devices are often deployed in tightly confined spaces. Thus, it is desirable for the multiplexing hardware be as compact as possible.

(5) FIG. 1 depicts an exemplary free-space WDM device **10** that uses a plurality of bandpass filters **12** (e.g., thin film filters) to define an optical path for an optical beam **14**. The optical beam **14** may include one or more optical signals **16** (e.g., n optical signals each having a different wavelength $\lambda_{\text{sub}.n}$). The WDM device **10** further includes a housing **18**, a common port collimator **20**, a plurality of channel port collimators **21**, an expansion port collimator **22**, and a tap port collimator **23**. An optical splitter **26** may be placed in the optical beam **14** to reflect a portion of the optical beam **14** (e.g., a portion having an amplitude 10 to 30 dB below that of the incident optical beam **14**) toward the tap port collimator **23**. Each collimator **20-23** has one end that faces toward the interior of the housing **18**, and another end that is operatively coupled a respective optical fiber **28-31**.

(6) The optical fiber **28** associated with the common port collimator **20** carries the optical beam **14**, and may be referred to as a “common optical fiber”. Each channel optical fiber **29** is configured to carry a respective portion of the optical beam **14** that includes an optical signal **16**, and is associated with a respective channel port collimator **21**. The optical fiber **30** associated with the expansion port collimator **22** may carry a portion of the optical beam **14** including one or more optical signals **16** not associated with any of the channel port collimators **21**. The optical fiber **31** associated with the tap port collimator **23** carries the tap signal, and may be used for testing during installation or maintenance, for example.

(7) Each filter **12** has a passband that passes a range of wavelengths that includes the wavelength $\lambda_{\text{sub}.n}$ of a respective one of the optical signals **16**, and excludes the wavelengths $\lambda_{\text{sub}.n}$ of the other optical signals **16**. Each filter **12** transmits the portion of the optical beam **14** including the

optical signal **16** having the wavelength $\lambda_{\text{sub},n}$ that falls within its passband, and reflects the portion of the optical beam **14** including the optical signals **16** having wavelengths $\lambda_{\text{sub},n}$ that fall outside its passband. By arranging the filters **12** sequentially in the optical path, each filter **12** can: (a) separate one of the optical signals **16** from the optical beam **14** and provide the separated optical signal **16** to a respective collimator **21** (demultiplexing application); and/or (b) add one of the optical signals **16** from the respective collimator **21** to the optical beam **14** (multiplexing application). Thus, WDM devices that use filters are typically bi-directional, meaning that they can both split optical signals received from an optical fiber for distribution at the network location, and combine optical signals for transmission into the optical fiber at the network location. Thus, WDM devices can be used as a multiplexer, demultiplexer, or both a multiplexer and demultiplexer.

(8) The optical components of WDM devices are typically assembled onto a rigid surface (e.g., an inner surface of the housing **18**) by an operator. To provide the operator with guidance on the position and orientation (or “placement”) of the components, the rigid surface may include visual alignment markings **34** that provide a visual indication of where each optical component should be placed. Together, position and orientation define the placement of the optical component within the WDM device **10**. To obtain optimal placement, i.e., a placement that optimizes performance of the WDM device, the operator may be required to actively position and orient, or “align”, each of the collimators **20-23** and the filters **12** during assembly.

(9) Due to stresses between the filtering coatings and anti-reflective coatings and the substrate of each filter **12**, the surfaces of the filters **12** are slightly curved. This curvature is stronger for the side having the filtering coatings than the side having anti-reflection coatings on the backside due to the higher number of layers in the filtering coatings. Characteristics of the filters include loss (both for the reflected and transmitted beams), bandwidth, and center wavelength. The center wavelength of a filter **12** defines the range of wavelengths that are transmitted by the filter and is optimized for a particular angle of incidence at which the filter **12** is to be used. Typical angles of incidence include 13.5 degrees for a Coarse Wavelength Division Multiplexing (CWDM), such as depicted by FIG. 1, and 1.8 degrees or 4 degrees for Dense Wavelength Division Multiplexing (DWDM). Other angles of incidence may also be possible for specific applications.

(10) Due to manufacturing tolerances and different coating designs (e.g., for different center wavelengths), the above described filter parameters show a certain degree of variation. For example, the center wavelength (which is purposely varied between filters **12** associated with different channels) may have an offset from the desired center wavelength. This center wavelength shift results in a difference between the designed “nominal” center wavelength and the measured “actual” center wavelength. The center wavelength is also dependent on the angle of incidence of the optical beam **14**, and may be tuned by changing the angle of the filter **12** relative to the optical beam **14**.

(11) By way of example, FIG. 2 depicts a graph showing the change in actual center wavelength for a filter **12** having a nominal center wavelength of 1550 nm when operating at a normal incidence (zero degrees) of the optical beam. When the angle of incidence is offset from the normal incidence by 1 degree, the actual center wavelength at which the filter **12** operates shifts to about 1549.9 nm, or about 0.10 nm shorter than the nominal center wavelength at normal incidence. As the offset in angle of incidence increases, the actual center wavelength becomes progressively shorter, and is about 0.45 nm shorter at 2 degrees, 1.00 nm shorter at 3 degrees, 1.85 nm shorter at 4 degrees, and about 2.88 nm shorter at 5 degrees. In operation, to avoid reflecting a portion of the optical beam back on itself, the designed angle of incidence is typically not zero degrees, but rather some non-zero value, e.g., 1.8, 4, or 13.5 degrees. Thus, the actual center wavelength of the filter **12** may be tuned from its nominal center wavelength to shorter or longer wavelengths by adjusting the orientation of the filter **12**.

(12) It is desirable to minimize the difference between the actual center wavelength of the filter and the wavelength of the optical signal **16** being passed by each filter **12** so that the WDM device **10**

has optimum spectral performance. Accordingly, the actual center wavelength may be adjusted so that it is aligned with the center wavelength of the optical signal **16**, which may be defined, for example, by the Telecommunication Standardization Sector of the International Telecommunication Union (ITU) in ITU-T G.694.1. To achieve this, the filters **12** may have to be arranged at angles different from the design angle of incidence to account for variations in the nominal center frequency of filters **12** from the design specification. Hence, active alignment of filters **12** is often needed to optimize performance of the WDM device **10**.

(13) FIG. 3 depicts an exemplary collimator **40** including a collimating lens **42** and a housing **44**. The housing **44** is configured to hold the collimating lens **42** in alignment with a ferrule **46**. The ferrule **46** includes a bore which holds an optical fiber **48**. The collimating lens **42** and ferrule **46** may be held in place within the housing **44** by an adhesive **50**, and the optical fiber **48** may be secured to the ferrule **46** by another adhesive **52**, which may be present within the bore of the ferrule **46** (not shown) and/or at a back end of the ferrule **46** (as shown in FIG. 3). The collimating lens **42** and ferrule **46** include respective interfaces **54**, **56** that are separated by an air gap **58**. The interfaces **54**, **56** are typically oriented at an angle (e.g., an eight degree angle) to avoid light reflected from the collimating lens **42** from being reflected back into the optical fiber **48**. The optical surfaces of the collimating lens **42** and optical fiber **48** may also be anti-reflection coated to minimize losses.

(14) The air gap **58** has a length $d_{sub,AG}$ that can be adjusted to define a working distance $d_{sub,WD}$ and an image distance $d_{sub,ID}$ of the collimator **40**. As used herein, the image distance $d_{sub,ID}$ refers to the distance from the collimating lens **42** to the beam waist **60**. The working distance $d_{sub,WD}$ refers to the distance at which the beam diameter behind the beam waist is equal to the diameter at the launch collimator. In the exemplary collimator **40** of FIG. 3, the working distance $d_{sub,WD}$ is twice the image distance $d_{sub,ID}$ on the output side of the collimator **40**. The length d_{AG} of the air gap **58** in each of the collimators **20-23** may be set to provide an optimum working distance $d_{sub,WD}$ for each collimator **20-23**, which may vary between collimators **20-23**.

(15) The parameters of the collimator **40** have a certain degree of variance that makes precise adjustments difficult. The bulk of this variance may be caused by the assembly tolerances of the collimator **40**, e.g., rotational misalignment between the ferrule **46** and the collimating lens **42**, lateral misalignment between the optical fiber **48** and the optical axis of the collimator lens **42**, or an offset of the length of the air gap $d_{sub,AG}$. These tolerances can lead to variations of the collimator working distance $d_{sub,WD}$ and pointing angle. Thus, the optical beam **14** is often not emitted straight from the collimator **40**, but at an angle, and the beam waist **60** is often not located at the desired position along the optical beam **14**. As a consequence, collimators **40** also have to be actively aligned for optimum optical coupling through the WDM device **10**.

(16) The active alignment of collimators and filters make the assembly of a compact module a complex and lengthy process. Having a large number of components that need alignment also increases the chances of a mis-aligned component, which may reduce the performance of the WDM device. Thus, there is a need in the fiber optics industry for improved devices and methods for assembling WDM devices with optimized performance.

SUMMARY

(17) In an aspect of the disclosure, an improved wavelength division multiplexing device is disclosed. The device includes a housing, a designed optical path at least partially contained within the housing, an alignment substrate associated with the housing, and a plurality of collimating lenses each having a collimator optical axis. The alignment substrate includes a plurality of aligners each configured to receive a respective collimating lens of the plurality of collimating lenses. When the respective collimating lens is received in the aligner, the collimator optical axis is aligned with the designed optical path.

(18) In an embodiment of the disclosed device, the alignment substrate may include an upper

surface, and each of the plurality of aligners may include a groove formed in the upper surface of the alignment substrate. By way of example, the groove may be V-shaped or U-shaped and sized to hold the collimating lens.

(19) In another embodiment of the disclosed device, the alignment substrate may further include a plurality of integrated optical waveguides each having a waveguide optical axis, and when the respective collimating lens is received in the aligner, the collimator optical axis may be aligned with the waveguide optical axis of a respective integrated optical waveguide of the plurality of integrated optical waveguides.

(20) In another embodiment of the disclosed device, the alignment substrate may further include a cavity in the upper surface that defines a lower surface, each groove formed in the upper surface of the alignment substrate may include an open end facing the cavity, and at least a portion of the designed optical path may pass through the cavity.

(21) In another embodiment of the disclosed device, the alignment substrate may include an upper surface, and each of the plurality of aligners may include one or more projections extending from the upper surface such that the collimator optical axis of the respective collimating lens is elevated above the upper surface of the alignment substrate.

(22) In another embodiment of the disclosed device, the device may further include a plurality of optical fibers each having an end face, and the respective collimating lens may be directly coupled to the end face of a respective optical fiber of the plurality of optical fibers. For example, the end face of the respective optical fiber may be directly coupled to the respective collimating lens by an adhesive or by welding the end face to the respective collimating lens.

(23) In another embodiment of the disclosed device, each of the plurality of collimating lenses may include an abutment, each of the plurality of aligners may include a stop, and each of the plurality of collimating lenses may be placed in a respective predetermined position along the designed optical path when the abutment of the collimating lens is in contact with the stop of the aligner.

(24) In another embodiment of the disclosed device, the device may further include at least one filter that defines an actual optical path for an optical beam, and the at least one filter may be configured to receive the optical beam from a first collimating lens of the plurality of collimating lenses, transmit at least a first portion of the optical beam toward a second collimating lens of the plurality of collimating lenses, and reflect at least a second portion of the optical beam toward a third collimating lens of the plurality of collimating lenses. In this embodiment, for example, the alignment substrate may further include one or more alignment markings that provide an indication of a placement of the at least one filter on the alignment substrate such that the actual optical path of the optical beam is aligned with the designed optical path.

(25) In another aspect of the disclosure, an improved method of making a wavelength division multiplexing device is disclosed. The method includes forming the alignment substrate having the plurality of aligners, each of the plurality of aligners configured to receive a respective collimating lens of the plurality of collimating lenses such that when the respective collimating lens is received in the respective aligner, the collimator optical axis is aligned with the designed optical path.

(26) In one embodiment, the method may further include forming the plurality of integrated optical waveguides in or on the alignment substrate each including the waveguide optical axis. Each of the aligners is configured to align the collimator optical axis of the respective collimating lens with both the designed optical path and the waveguide optical axis of the respective integrated optical waveguide of the plurality of integrated optical waveguides.

(27) In another aspect of the disclosure, another improved method of making a wavelength division multiplexing device is disclosed and includes providing the alignment substrate having the plurality of aligners, each of the plurality of aligners configured to receive a respective collimating lens of the plurality of collimating lenses such that when the respective collimating lens is received in the respective aligner, the collimator optical axis is aligned with the designed optical path, and placing each of the plurality of collimating lenses in a respective aligner of the plurality of aligners.

(28) In another embodiment of the disclosed method, the alignment substrate may include the plurality of integrated optical waveguides in or on the alignment substrate each including the waveguide optical axis, and placing each of the plurality of collimating lenses in the respective aligner may align the collimator optical axis of the respective collimating lens with the waveguide optical axis of a respective integrated optical waveguide of the plurality of integrated optical waveguides.

(29) In another embodiment of the disclosed method, each collimating lens of the plurality of collimating lenses may include the abutment, each aligner of the plurality of aligners may include the stop, and the method may further include abutting the abutment of the respective collimating lens with the stop of the respective aligner. Abutting the abutment with the stop may position the respective collimating lens along the designed optical path.

(30) In another embodiment of the disclosed method, the method may further include placing at least one filter that defines the actual optical path for the optical beam on the alignment substrate. The at least one filter may be configured to receive the optical beam from the first collimating lens of the plurality of collimating lenses, transmit at least the first portion of the optical beam toward the second collimating lens of the plurality of collimating lenses, and reflect at least the second portion of the optical beam toward the third collimating lens of the plurality of collimating lenses. In this embodiment, the alignment substrate may further include one or more alignment markings that provide the indication of the placement of the at least one filter on the alignment substrate such that the actual optical path of the optical beam is aligned with the designed optical path, and the method may further include aligning the at least one filter with a respective alignment mark of the one or more alignment marks.

(31) In another embodiment of the disclosed method, the alignment substrate may further include the cavity in the upper surface that defines the lower surface, and at least a portion of the designed optical path may pass through the cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiment(s), and together with the description serve to explain principles and operation of the various embodiments. Features and attributes associated with any of the embodiments shown or described may be applied to other embodiments shown, described, or appreciated based on this disclosure.

(2) FIG. 1 is a top view of an exemplary WDM device including a plurality of filters and collimators;

(3) FIG. 2 is a graphical view depicting changes in the center frequency of a filter of the WDM device of FIG. 1 in response to different angular offsets from the normal angle of incidence;

(4) FIG. 3 is a diagrammatic cross-sectional view of a collimator of the WDM device of FIG. 1;

(5) FIG. 4 is a diagrammatic cross-sectional view of another type of collimator that may be used in a WDM device which includes an alignment substrate;

(6) FIG. 5 is a diagrammatic view of an exemplary alignment substrate for a WDM device;

(7) FIG. 6 is a diagrammatic view of a WDM device including the alignment substrate of FIG. 5;

(8) FIG. 7 is a diagrammatic view of a WDM device including another exemplary alignment substrate;

(9) FIG. 8 is a diagrammatic view of another exemplary alignment substrate including a plurality of integrated optical waveguides and alignment grooves;

(10) FIG. 9 is a diagrammatic view of a WDM device including the alignment substrate of FIG. 8;

- (11) FIG. 10 is a perspective view of a portion of the alignment substrate of FIG. 8;
- (12) FIGS. 11-14 are cross-sectional views of various exemplary grooves that may be used in the alignment substrates of FIGS. 5 and 8;
- (13) FIGS. 15A-17D are perspective views of various exemplary alignment features that may be used to provide aligners in the alignment substrates of FIGS. 5 and 8.

DETAILED DESCRIPTION

- (14) Various embodiments will be further clarified by examples in the description below. In general, the description relates to WDM devices that include a substrate configured to passively align one or more optical components of the WDM device. The WDM devices may also use collimators that all have the same fixed working distance, e.g., $d_{\text{sub.WD}}=0$. Eliminating the need for an adjustable working distance may allow elimination of the air gap 58 applied in conventional collimators 40, such as depicted by FIG. 3. This may allow collimators to be reduced to only two components, a fiber and a collimating lens that transforms the diameter of the optical beam, or in some embodiments, just the collimating lens. The diameter of the optical beam may be adjusted such that the focusing effect of the curved surface of the filters compensates for diffractive beam widening.
- (15) FIG. 4 depicts an exemplary collimator 70 (sometimes referred to as a “fiber collimator” or “pigtail collimator”) including a collimating lens 72 operatively coupled to an optical fiber 74. The collimating lens 72 includes an optical axis 75, a cylindrical surface 76, a waveguide interface surface 78, and an air interface surface 80. The waveguide interface surface 78 is joined to the air interface surface 80 by the cylindrical surface 76, and oriented at a generally orthogonal angle relative to the optical fiber 74. One end of the optical fiber 74 may be directly coupled to the waveguide interface surface 78 of collimating lens 72 by an adhesive 82 (shown) or by welding/fusing an end of the optical fiber 74 to the waveguide interface surface 78, e.g., by laser processing or an electric arc. The collimating lens 72 may be configured (e.g., by choice of refractive index, length, and radius of curvature of the air interface surface 80) so that the waist 84 of an optical beam 86 emitted by the collimator 70 is located at or proximate to the air interface surface 80. This may result in the collimator 70 having a working distance $d_{\text{sub.WD}}=0$.
- (16) Eliminating the air gap between the collimating lens 72 and optical fiber 74 may avoid the need for a housing or ferrule. Directly coupling the collimating lens 72 to the optical fiber 74 may also produce a more accurate alignment between the optical axis 75 of collimating lens 72 and the optical fiber 74 associated with the collimating lens 72. Eliminating the collimator housing may also avoid introducing mechanical tolerances that can be a source of misalignments between the collimating lens 72 and the other optical components of the WDM device. This improvement in alignment may be due to the orientation of the collimating lens 72 within the WDM device being defined by the cylindrical surface 76 of the collimating lens 72 rather than by, with reference to FIG. 3, how accurately the collimating lens 42 and ferrule 46 are aligned with each other as well as the housing 44. Both the direct coupling of the collimating lens 72 to the optical fiber 74 and the elimination of the collimator housing 44 may reduce pointing angle misalignment as compared to the collimator 40 depicted by FIG. 3. Elimination of the collimator housing 44 may also facilitate alignment between the collimating lens 72 and one or more passive alignment features, as described in detail below.
- (17) Advantageously, the exemplary collimator 70 provides a simplified collimator structure that contributes to improved collimator quality and accuracy by avoiding the variations introduced by use of an air gap, housing, ferrule, and angled interfaces. The exemplary collimator 70 also avoids the need for anti-reflection coatings on the end-faces of the collimating lens 72 and optical fiber 74, naturally provides improved alignment between the optical fiber 74 and collimating lens 72, reduces pointing angle errors, working distance offsets, and lowers the manufacturing cost of the collimator 70. Once a relation between the cylindrical surface 76 of collimating lens 72 and the direction of the optical beam 86 is determined, alignment features may be defined with

predetermined positions and orientations on a substrate such that each of the collimators **70** will be properly aligned upon assembly of the WDM device.

(18) FIGS. **5** and **6** provide top-views depicting an alignment substrate **90** (FIG. **5**) and an exemplary free-space three-port WDM device **92** (FIG. **6**). The WDM device **92** includes a filter **12**, the alignment substrate **90**, a common port collimator **96** that functions as a common port, a transmission port collimator **97** that functions as a transmission port, and a reflection port collimator **98** that functions as a reflection port. The alignment substrate **90** includes an upper surface **100**, a cavity **102** that defines a lower surface **104** offset from the upper surface **100**, and a plurality of alignment features (“aligners”) **106** each comprising a groove (e.g., a V-shaped groove) having a distal end open to the cavity **102**. The lower surface **104** may include visual alignment markings **108** indicative of a placement for the filter **12**. Each aligner **106** may be configured to receive the collimating lens **72** of a respective collimator **96-98**, and to align each collimator **96-98** so that the optical beam **86** follows a designed optical path **109** between the filter **12** and collimators **96-98**. The aligners **106** may be fabricated, for example, using photolithography, high-precision milling, laser processing, or any other suitable technique.

(19) The WDM device **92** represents a WDM device according to this disclosure in its simplest form. When operating in a demultiplexing mode, the optical beam **86** is launched from the common port collimator **96** towards the filter **12**. A portion of the optical beam **86** may be transmitted by the filter **12** and received by the transmit port collimator **97**. The remainder of the optical beam **86** may be reflected toward and received by the reflection port collimator **98**. Both the collimators **96-98** and the aligners **106** can be made with high precision, so that only the filter **12** requires active alignment.

(20) The optimal position and orientation of the filter **12** may be driven by the positions of the common and reflection port collimators **96, 98**. However, variations of the surface curvature of the filters **12** may require active alignment of each filter **12**. In cases where the position of the reflection port is fixed (such as depicted by FIG. **6**), it may only be possible to optimize one of insertion loss or center wavelength shift at a time through alignment of the filter **12**. In this type of WDM device, a compromise may be found that minimizes the impact on overall performance by balancing insertion loss and center wavelength shift. This type of filter arrangement may therefore be more suitable for CWDM than for DWDM since the filter bandwidth is much wider for CWDM, e.g., about 0.7 nm for DWDM vs. 20 nm for CWDM filters at full-width half-maximum. CWDM devices may therefore be more tolerant of center wavelength shift. To improve performance (e.g., for DWDM applications), an additional adjustable optical element such as a wedge or a lens (not shown) may be inserted into the optical path to provide an additional degree of freedom to the design.

(21) Manufacturing tolerances of the filters **12** may be compensated for by active alignment of the filters **12**. To optimize coupling between the optical beam **86** and the optical fiber **74** on the reflection port, the filter **12** may be moved laterally (as depicted by double arrowed line **88**) and along the optical beam **86** (left-right as depicted by double arrowed line **89**) without affecting the coupling between the optical beam **86** and the optical fiber **74** on the transmission port. Any rotation of the filter **12** may change the lateral shift position of the optical beam **86** in the plane of the transmission port, which can affect coupling between the optical beam **86** and optical fiber **74** at the transmission port.

(22) FIG. **7** depicts an exemplary free-space multi-port device **110** including a plurality of filters **12**, collimators **70**, a housing **111**, and an alignment substrate **112** associated with the housing **111** and having a plurality of aligners **106** and visual alignment markings **108**. The alignment substrate **112** may be associated with the housing **111**, for example, by operatively coupling the alignment substrate **112** to the housing **111** (e.g., using fasteners or adhesive), or by forming alignment substrate **112** as an integral part of the housing **111** (e.g., by machining or molding the housing **111** and alignment substrate **112** as a single unit). For WDM devices in which the filters **12** are angled

(such as depicted by FIGS. 6 and 7), the aligners **106** for the common and transmission port collimators **70** may be laterally shifted with respect to each other as the optical beam **86** experiences a lateral shift by refraction at the surfaces of the filters **12**.

(23) For a WDM device in which all the collimators have a working distance $d_{sub.WD}=0$, the performance of the WDM device may be relatively tolerant to collimating lens alignment errors along the length of the optical path. That is, only the lateral position of the collimator lenses **72** may require high precision, which may be provided, for example, by linear indentations into the substrate (e.g., V-shaped grooves) or projections (e.g., parallel walls or linearly spaced pylons). High precision structures can be fabricated using photolithography, high-precision milling, laser processing, or any other suitable technique.

(24) FIGS. 8 and 9 provide top-views depicting an exemplary alignment substrate **120** (FIG. 8) and free-space multi-port WDM device **122** (FIG. 9). The WDM device **122** includes a plurality of filters **12**, a plurality of collimating lenses **72**, a housing **124** associated with the alignment substrate **120**, and a plurality of integrated optical waveguides **126**. Each integrated optical waveguide **126** may be configured to couple an optical beam comprising one or more optical signals between an optical connector **128**, **130** and a respective collimating lens **72**. The integrated optical waveguides **126** may be fabricated, for example, using a multi-step process including one or more sub-processes, such as deposition, photolithography, ion exchange, etching, or laser writing, or by any other suitable means. The alignment substrate **120** may include an upper surface **132**, a cavity **134** that defines a lower surface **136** offset from the upper surface **132**, and a plurality of aligners **138**. Each aligner **138** may comprise a groove (e.g., a V-shaped groove) having a distal end open to the cavity **134**.

(25) FIG. 10 is a perspective view of a portion of the alignment substrate **120** showing details of exemplary aligners **138**. Each aligner **138** comprises a groove **140** including diagonal surfaces **142** arranged in a V-shape, an open end **144** facing the cavity **134**, and a closed end **146** opposite the open end **144**. The groove **140** may have a length such that, when the waveguide interface surface **78** of collimating lens **72** abuts the closed end **146** of groove **140**, the collimating lens **72** is aligned linearly along the optical path of the WDM device **122**. Thus, the interface surface **78** of collimating lens **72** may provide an abutment **147** and the closed end **146** of groove **140** may provide a stop **149**. The abutment **147** and stop **149** may cooperate to position the collimating lens **72** along the designed optical path **109** when the abutment **147** of the collimating lens **72** is in contact with the stop **149** of the aligner **138**. The diagonal surfaces **142** of the groove **140** are configured to position the collimating lens **72** so that its optical axis **75** is aligned laterally (vertically and horizontally) and angularly with both the end face of the integrated optical waveguide **126** associated with the aligner **138** and the optical path of WDM device **122**. This placement may be achieved through contact between the diagonal surfaces **142** of groove **140** and the cylindrical surface **76** of collimating lens **72**.

(26) FIGS. 11 and 12 provide cross-sectional views of the V-shaped groove **140** and an exemplary stepped groove **148**. The stepped groove **148** includes steps **150** each having a corner **152** that supports the culminating lens **72**. Each of the grooves **140**, **148** is configured to position the collimating lens **72** vertically and horizontally through contact with its cylindrical surface **76** such that the optical axis **75** of the collimating lens **72** is aligned with both the integrated optical waveguide **126** and the optical path. This alignment may be accomplished by configuring each groove **140**, **148** to contact the cylindrical surface **76** of collimating lens **72** along two points of contact, e.g., lines of contact with the diagonal surfaces **142** of groove **140** or the corners **152** of each step **150** of groove **148**.

(27) FIGS. 13 and 14 provide cross-sectional views of another V-shaped groove **154** and an exemplary raised groove **156**. The raised groove **156** may comprise a pair of parallel steps **158** that project upward from the upper surface **132** of alignment substrate **120**. Each of the grooves **154**, **156** is configured to position the collimating lens **72** vertically and horizontally through contact

with its cylindrical surface **76** so that the optical axis **75** of the collimating lens **72** is above the upper surface **132** of alignment substrate **120**. This may provide clearance to accommodate the optical fiber **74** for embodiments in which the optical fiber **74** is directly coupled to the waveguide interface surface **78** of collimating lens **72**. This raised vertical positioning of the collimating lenses **72** may also allow omission of the cavity **102** by raising the optical path of the WDM device **122** above the upper surface **132** of alignment substrate **120**.

(28) FIGS. **15A** and **15B** are perspective views of a portion of two embodiments of an alignment substrate **160** including a top surface **162** and exemplary aligners **164** each including the raised groove **156** of FIG. **14**. Each aligner **164** may include a pair of parallel steps **158**, with the embodiment depicted by FIG. **15A** also including a cross-step **166** that joins the proximal ends of the parallel steps **158**. When present, the cross-step **166** may provide a stop **168** that aligns the collimating lens **72** along the predetermined optical path of the optical beam **86**. Each parallel step **158** may provide a point of contact **169** in the form of an edge that contacts the cylindrical surface **76** of collimating lens **72**.

(29) FIG. **16** is a perspective view of a portion of an alignment substrate **170** including a top surface **172** and exemplary aligners **174** having a similar configuration as the aligners **164** of FIG. **15A**, but with a portion of each parallel step **158** between the proximal and distal ends thereof omitted. The resulting aligner **174** includes a pair of forward supports **176** and a rear support **178**. Each forward support **176** may comprise a column that projects upward from the top surface **172** of alignment substrate **170**, and which provides a point of contact **180** with the cylindrical surface **76** of collimating lens **72**. The forward supports **176** may thereby cradle a portion of the cylindrical surface **76** of collimating lens **72** proximate the air interface surface **80** thereof. The rear support **178** may include the cross-step **166** of FIG. **15A** to align the collimating lens **72** along the optical path, and projections **182** each providing a point of contact **184** with the cylindrical surface **76** of collimating lens **72**. The projections **182** may operate cooperatively with the forward supports **176** to define the orientation of the collimating lens **72** such that the optical axis **75** thereof is aligned with the optical path.

(30) FIGS. **17A-17D** are perspective views of a portion of an alignment substrate **190** including a top surface **192** and exemplary aligners **194**. Each aligner **194** includes a plurality of supports, including the forward supports **176** and rear supports **196**. Each rear support **196** comprises a column that projects upward from the top surface **192** of alignment substrate **190** and which provides a point of contact **202** with the cylindrical surface **76** of collimating lens **72**. The embodiments depicted by FIGS. **17A-17C** also include one or more stops **198**. Each stop **198** provides a surface **200** which aligns the collimating lens **72** along the predetermined optical path. The stop **198** of FIG. **17B** includes a groove **204** (e.g., a U-shaped groove) to accommodate the optical fiber **74** and/or the adhesive **82** depicted by FIG. **4**. The stops **198** depicted by FIG. **17C** are arranged to provide a gap **206** between the stops **198** to accommodate the optical fiber **74** and/or adhesive **82**.

(31) Advantages provided by aligner substrates may include simplification of the free-space WDM device assembly (thereby reducing the required assembly time), elimination of the need for active collimator alignment, lower WDM device cost, reduced component cost (e.g., simplified collimators), reduced WDM device size, and elimination of the need to accommodate optical fiber bend radiuses for embodiments using integrated optical waveguides **126**.

(32) While the present disclosure has been illustrated by the description of specific embodiments thereof, and while the embodiments have been described in considerable detail, it is not intended to restrict or in any way limit the scope of the appended claims to such detail. The various features discussed herein may be used alone or in any combination within and between the various embodiments. Additional advantages and modifications will readily appear to those skilled in the art. The present disclosure in its broader aspects is therefore not limited to the specific details, representative apparatus and methods and illustrative examples shown and described. Accordingly,

departures may be made from such details without departing from the scope of the present disclosure.

Claims

1. A method of making a wavelength division multiplexing device including a plurality of collimating lenses each having a collimator optical axis aligned with a designed optical path, the method comprising: forming an alignment substrate having a plurality of aligners, each of the plurality of aligners configured to receive a respective collimating lens of the plurality of collimating lenses such that when the respective collimating lens is received in a respective aligner, the collimator optical axis is aligned with the designed optical path, wherein each collimating lens of the plurality of collimating lenses includes an abutment and each aligner of the plurality of aligners includes a stop; and abutting the abutment of the respective collimating lens with the stop of the respective aligner, wherein abutting the abutment with the stop positions the respective collimating lens along the designed optical path, wherein, for each of the plurality of collimating lenses, the abutment is provided at an end of the respective collimating lens which is configured to be coupled to an optical fiber.
2. The method of claim 1, further comprising: forming a plurality of integrated optical waveguides in or on the alignment substrate each including a waveguide optical axis, and wherein each of the aligners is configured to align the collimator optical axis of the respective collimating lens with the waveguide optical axis of a respective integrated optical waveguide of the plurality of integrated optical waveguides.
3. The method of claim 1, wherein the alignment substrate further includes a cavity in the upper surface that defines a lower surface, and at least a portion of the designed optical path passes through the cavity.
4. A method of making a wavelength division multiplexing device including a plurality of collimating lenses each having a collimator optical axis aligned with a designed optical path, the method comprising: providing an alignment substrate having a plurality of aligners, each of the plurality of aligners configured to receive a respective collimating lens of the plurality of collimating lenses such that when the respective collimating lens is received in a respective aligner, the collimator optical axis is aligned with the designed optical path; placing each of the plurality of collimating lenses in a respective aligner of the plurality of aligners, wherein each collimating lens of the plurality of collimating lenses includes an abutment and each aligner of the plurality of aligners includes a stop; and abutting the abutment of the respective collimating lens with the stop of the respective aligner, wherein abutting the abutment with the stop positions the respective collimating lens along the designed optical path, wherein, for each of the plurality of collimating lenses, the abutment is provided at an end of the respective collimating lens which is configured to be coupled to an optical fiber.
5. The method of claim 4, wherein: the alignment substrate includes a plurality of integrated optical waveguides in or on the alignment substrate each including a waveguide optical axis, and placing each of the plurality of collimating lenses in the respective aligner aligns the collimator optical axis of the respective collimating lens with the waveguide optical axis of a respective integrated optical waveguide of the plurality of integrated optical waveguides.
6. The method of claim 4, further comprising: placing at least one filter that defines an actual optical path for an optical beam on the alignment substrate, wherein the at least one filter is configured to receive the optical beam from a first collimating lens of the plurality of collimating lenses, transmit at least a first portion of the optical beam toward a second collimating lens of the plurality of collimating lenses, and reflect at least a second portion of the optical beam toward a third collimating lens of the plurality of collimating lenses.
7. The method of claim 6, wherein the alignment substrate further includes one or more alignment

- markings that provide an indication of a placement of the at least one filter on the alignment substrate such that the actual optical path of the optical beam is aligned with the designed optical path, and further comprising: aligning the at least one filter with a respective alignment mark of the one or more alignment marks.
8. A method of making a wavelength division multiplexing device including a plurality of collimating lenses each having a collimator optical axis aligned with a designed optical path, the method comprising: providing an alignment substrate having a plurality of aligners, each of the plurality of aligners configured to receive a respective collimating lens of the plurality of collimating lenses; placing each of the plurality of collimating lenses in a respective aligner of the plurality of aligners, wherein each collimating lens of the plurality of collimating lenses includes an abutment and each aligner of the plurality of aligners includes a stop; and abutting the abutment of the respective collimating lens with the stop of the respective aligner, wherein abutting the abutment with the stop positions the respective collimating lens along the designed optical path, wherein, for each of the plurality of collimating lenses, the abutment is provided at an end of the respective collimating lens which is configured to be coupled to an optical fiber.
9. The method of claim 8, wherein: the alignment substrate includes a plurality of integrated optical waveguides in or on the alignment substrate each including a waveguide optical axis, and placing each of the plurality of collimating lenses in the respective aligner aligns the collimator optical axis of the respective collimating lens with the waveguide optical axis of a respective integrated optical waveguide of the plurality of integrated optical waveguides.
10. The method of claim 8, further comprising: placing at least one filter that defines an actual optical path for an optical beam on the alignment substrate, wherein the at least one filter is configured to receive the optical beam from a first collimating lens of the plurality of collimating lenses, transmit at least a first portion of the optical beam toward a second collimating lens of the plurality of collimating lenses, and reflect at least a second portion of the optical beam toward a third collimating lens of the plurality of collimating lenses.
11. The method of claim 10, wherein the alignment substrate further includes one or more alignment markings that provide an indication of a placement of the at least one filter on the alignment substrate such that the actual optical path of the optical beam is aligned with the designed optical path, and further comprising: aligning the at least one filter with a respective alignment mark of the one or more alignment marks.
12. The method of claim 8, wherein the alignment substrate further includes a cavity in the upper surface that defines a lower surface, and at least a portion of the designed optical path passes through the cavity.
13. The method of claim 8, wherein each of the plurality of aligners is configured to receive a respective collimating lens of the plurality of collimating lenses such that when the respective collimating lens is received in a respective aligner, the collimator optical axis is aligned with the designed optical path.
14. The method of claim 8, wherein each of the plurality of collimating lenses has a collimator axis that is aligned with the designed optical path after the collimating lens is placed in the respective aligner.
15. The method of claim 8, further comprising the step of placing the alignment substrate at least partially within a housing such that the designed optical path is at least partially contained within the housing.
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